Android Application Development Lab Project

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School of Computer Applications

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Android Application Development Lab Project.....1

Java Code

Activity Main :

```
public class MainActivity extends AppCompatActivity {

  // UI elements
  private LinearLayout taskContainer;
  private View addBtn;
  private EditText searchBar;
  private TextView titleTv;

  // Application state
  private String userName = "User";
  private String selectedCategory = "All";

  // Data model for a single Todo task
  static class Task {
     String title, desc, color, category;
     Long dueDate;
     boolean isCompleted;

     Task(String title, String desc, String color, String category, Long dueDate, boolean isCompleted)
{
        this.title = title;
        this.desc = desc;
        this.color = color;
        this.category = category;
        this.dueDate = dueDate;
        this.isCompleted = isCompleted;
     }
  }

  // List to store task objects and constants for persistence
  private final List<Task> taskList = new ArrayList<>();
  private static final String PREFS_NAME = "TasksPrefs";
  private static final String TASKS_KEY = "tasks";
  private static final String USER_NAME_KEY = "user_name";
  private static final String IS_REGISTERED_KEY = "is_registered";

  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);
```

```
        // Initialize UI components
        taskContainer = findViewById(R.id.taskContainer);
        addBtn = findViewById(R.id.addBtn);
        searchBar = findViewById(R.id.search);
        titleTv = findViewById(R.id.title);
        View resetBtn = findViewById(R.id.resetBtn);

        // Handle reset button click
        resetBtn.setOnClickListener(v -> showResetConfirmDialog());

        // Load existing data
        loadUserData();
        loadTasks();

        // Add a welcome task for first-time users if list is empty
        if (taskList.isEmpty()) {
            Task t1 = addTask("Welcome " + userName + "!", "Try adding some tasks.", "#E6D9FF",
"Personal", null, false);
            taskList.add(t1);
            saveTasks();
        }

        // Setup button listeners
        addBtn.setOnClickListener(v -> showAddDialog());
        titleTv.setOnClickListener(v -> showNameDialog());

        // Initialize features
        setupSearch();
        setupTabs();
        updateTitle();
    }

    // Shows a confirmation dialog before resetting all app data
    private void showResetConfirmDialog() {
        new AlertDialog.Builder(this)
                .setTitle("Reset App")
                .setMessage("Are you sure you want to reset the app? This will delete all your data.")
                .setPositiveButton("Reset", (dialog, which) -> {
                    SharedPreferences prefs = getSharedPreferences(PREFS_NAME, MODE_PRIVATE);
                    prefs.edit().clear().apply();

                    // Restart app to registration screen
                    android.content.Intent intent = new android.content.Intent(this, RegisterActivity.class);
                    intent.addFlags(android.content.Intent.FLAG_ACTIVITY_CLEAR_TOP |
android.content.Intent.FLAG_ACTIVITY_NEW_TASK);
                    startActivity(intent);
                    finish();
```

```
            })
            .setNegativeButton(R.string.cancel, null)
            .show();
    }

    // Loads the user's name from persistent storage
    private void loadUserData() {
        SharedPreferences prefs = getSharedPreferences(PREFS_NAME, MODE_PRIVATE);
        userName = prefs.getString(USER_NAME_KEY, "User");
    }

    // Updates the screen title with the user's name
    private void updateTitle() {
        titleTv.setText(getString(R.string.user_name_placeholder, userName));
    }

    // Shows a dialog to allow user to change their name
    private void showNameDialog() {
        EditText input = new EditText(this);
        input.setHint(R.string.enter_name_hint);
        input.setText(userName);

        new AlertDialog.Builder(this)
            .setTitle("Change Name")
            .setView(input)
            .setPositiveButton(R.string.save, (dialog, which) -> {
                userName = input.getText().toString();
                getSharedPreferences(PREFS_NAME,
MODE_PRIVATE).edit().putString(USER_NAME_KEY, userName).apply();
                updateTitle();
            })
            .setNegativeButton(R.string.cancel, null)
            .show();
    }

    // Sets up the search bar to filter tasks as the user types
    private void setupSearch() {
        searchBar.addTextChangedListener(new TextWatcher() {
            @Override
            public void beforeTextChanged(CharSequence s, int start, int count, int after) {}
            @Override
            public void onTextChanged(CharSequence s, int start, int before, int count) {
                filterTasks();
            }
            @Override
            public void afterTextChanged(Editable s) {}
        });
    }
```

```
    // Sets up click listeners for the category tabs
    private void setupTabs() {
        findViewById(R.id.tabAll).setOnClickListener(v -> selectTab("All"));
        findViewById(R.id.tabPersonal).setOnClickListener(v -> selectTab("Personal"));
        findViewById(R.id.tabWork).setOnClickListener(v -> selectTab("Work"));
        findViewById(R.id.tabShopping).setOnClickListener(v -> selectTab("Shopping"));
        findViewById(R.id.tabIdeas).setOnClickListener(v -> selectTab("Ideas"));

        selectTab("All");
    }

    // Handles category selection and updates tab UI styling
    private void selectTab(String category) {
        selectedCategory = category;

        LinearLayout tabs = findViewById(R.id.tabs);
        for (int i = 0; i < tabs.getChildCount(); i++) {
            View child = tabs.getChildAt(i);
            if (child instanceof TextView) {
                TextView tab = (TextView) child;
                if (tab.getText().toString().equalsIgnoreCase(category)) {
                    tab.setTextColor(ContextCompat.getColor(this, R.color.purple_primary));
                    tab.setBackgroundResource(R.drawable.tab_indicator);
                } else {
                    tab.setTextColor(ContextCompat.getColor(this, R.color.gray_text));
                    tab.setBackground(null);
                }
            }
        }

        filterTasks();
    }

    // Filters the task list based on search query and selected category
    private void filterTasks() {
        String query = searchBar.getText().toString().toLowerCase();
        for (int i = 0; i < taskContainer.getChildCount(); i++) {
            View view = taskContainer.getChildAt(i);
            Task task = (Task) view.getTag();

            boolean matchesQuery = task.title.toLowerCase().contains(query) ||
task.desc.toLowerCase().contains(query);
            boolean matchesCategory = selectedCategory.equals("All") ||
task.category.equalsIgnoreCase(selectedCategory);

            view.setVisibility(matchesQuery && matchesCategory ? View.VISIBLE : View.GONE);
        }
```

```
    }

    // Inflates a task UI item and adds it to the container
    private Task addTask(String title, String desc, String color, String category, Long dueDate, boolean
isCompleted) {
        final Task currentTask = new Task(title, desc, color, category, dueDate, isCompleted);

        View view = LayoutInflater.from(this).inflate(R.layout.task_item, taskContainer, false);
        ((androidx.cardview.widget.CardView)view).setCardBackgroundColor(Color.parseColor(color));

        // Get view references within task item
        TextView t = view.findViewById(R.id.taskTitle);
        TextView d = view.findViewById(R.id.taskDesc);
        TextView dateTv = view.findViewById(R.id.taskDueDate);
        TextView catTv = view.findViewById(R.id.taskCategory);
        CheckBox checkBox = view.findViewById(R.id.taskCheckbox);
        ImageButton menuBtn = view.findViewById(R.id.menuBtn);
        View dateSection = view.findViewById(R.id.dateSection);

        // Populate view data
        t.setText(title);
        d.setText(desc);
        catTv.setText("#" + category);
        checkBox.setChecked(isCompleted);

        updateTaskStyle(t, d, dateTv, isCompleted);

        // Handle date display if present
        if (dueDate != null) {
            Calendar cal = Calendar.getInstance();
            cal.setTimeInMillis(dueDate);
            String dateStr = cal.get(Calendar.DAY_OF_MONTH) + "/" + (cal.get(Calendar.MONTH) + 1)
+ "/" + cal.get(Calendar.YEAR);
            dateTv.setText(dateStr);
            dateSection.setVisibility(View.VISIBLE);
        }

        view.setTag(currentTask);

        // Handle completion toggle
        checkBox.setOnCheckedChangeListener((buttonView, isChecked) -> {
            currentTask.isCompleted = isChecked;
            updateTaskStyle(t, d, dateTv, isChecked);
            saveTasks();
        });

        // Handle task options menu (Delete)
        menuBtn.setOnClickListener(v -> {
```

```
            PopupMenu popup = new PopupMenu(this, v);
            popup.getMenu().add(0, 1, 0, R.string.menu_delete);
            popup.setOnMenuItemClickListener(item -> {
                if (item.getItemId() == 1) {
                    taskContainer.removeView(view);
                    taskList.remove(currentTask);
                    saveTasks();
                    return true;
                }
                return false;
            });
            popup.show();
        });

        taskContainer.addView(view);
        return currentTask;
    }

    // Updates text styling (like strike-through) based on task completion status
    private void updateTaskStyle(TextView t, TextView d, TextView dateTv, boolean isCompleted) {
        if (isCompleted) {
            t.setPaintFlags(t.getPaintFlags() | Paint.STRIKE_THRU_TEXT_FLAG);
            d.setPaintFlags(d.getPaintFlags() | Paint.STRIKE_THRU_TEXT_FLAG);
            dateTv.setPaintFlags(dateTv.getPaintFlags() | Paint.STRIKE_THRU_TEXT_FLAG);
            t.setTextColor(Color.GRAY);
        } else {
            t.setPaintFlags(t.getPaintFlags() & (~Paint.STRIKE_THRU_TEXT_FLAG));
            d.setPaintFlags(d.getPaintFlags() & (~Paint.STRIKE_THRU_TEXT_FLAG));
            dateTv.setPaintFlags(dateTv.getPaintFlags() & (~Paint.STRIKE_THRU_TEXT_FLAG));
            t.setTextColor(Color.BLACK);
        }
    }

    // Displays a dialog to create a new task
    private void showAddDialog() {
        View view = LayoutInflater.from(this).inflate(R.layout.dialog_add, null);
        EditText title = view.findViewById(R.id.newTitle);
        EditText desc = view.findViewById(R.id.newDesc);

        // Setup category spinner
        Spinner categorySpinner = new Spinner(this);
        String[] categories = {"Personal", "Work", "Shopping", "Ideas"};
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_spinner_item, categories);
        adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
        categorySpinner.setAdapter(adapter);
        categorySpinner.setPadding(0, 20, 0, 20);
```

```
    ((ViewGroup)view).addView(categorySpinner);

    // Date selection logic
    final Long[] selectedDate = {null};
    TextView selectedDateText = view.findViewById(R.id.selectedDateText);

    view.findViewById(R.id.pickDateBtn).setOnClickListener(v -> {
        Calendar cal = Calendar.getInstance();
        DatePickerDialog datePicker = new DatePickerDialog(this, (view1, year, month, dayOfMonth)
-> {
            Calendar selected = Calendar.getInstance();
            selected.set(year, month, dayOfMonth);
            selectedDate[0] = selected.getTimeInMillis();

            String dateStr = dayOfMonth + "/" + (month + 1) + "/" + year;
            selectedDateText.setText(getString(R.string.due_date_label, dateStr));
            selectedDateText.setVisibility(View.VISIBLE);
        }, cal.get(Calendar.YEAR), cal.get(Calendar.MONTH),
cal.get(Calendar.DAY_OF_MONTH));

        // Restrict past dates
        datePicker.getDatePicker().setMinDate(System.currentTimeMillis() - 1000);
        datePicker.show();
    });

    // Dialog buttons
    new AlertDialog.Builder(this)
            .setTitle(R.string.add_task_dialog_title)
            .setView(view)
            .setPositiveButton(R.string.save, (dialog, which) -> {
                String t = title.getText().toString();
                String d = desc.getText().toString();
                String cat = categorySpinner.getSelectedItem().toString();

                // Assign colors based on category
                String color = "#E6D9FF";
                if (cat.equals("Work")) color = "#E6D9FF";
                else if (cat.equals("Personal")) color = "#D9E8FF";
                else if (cat.equals("Shopping")) color = "#DFF5E1";
                else if (cat.equals("Ideas")) color = "#FFF0E0";

                if (!t.isEmpty()) {
                    Task newTask = addTask(t, d, color, cat, selectedDate[0], false);
                    taskList.add(newTask);
                    saveTasks();
                    filterTasks();
                }
            })
```

```
            .setNegativeButton(R.string.cancel, null)
            .show();
    }

    // Saves the task list to SharedPreferences as a JSON string
    private void saveTasks() {
        SharedPreferences prefs = getSharedPreferences(PREFS_NAME, MODE_PRIVATE);
        JSONArray jsonArray = new JSONArray();
        try {
            for (Task task : taskList) {
                JSONObject obj = new JSONObject();
                obj.put("title", task.title);
                obj.put("desc", task.desc);
                obj.put("color", task.color);
                obj.put("category", task.category);
                obj.put("dueDate", task.dueDate);
                obj.put("isCompleted", task.isCompleted);
                jsonArray.put(obj);
            }
        } catch (JSONException e) { e.printStackTrace(); }
        prefs.edit().putString(TASKS_KEY, jsonArray.toString()).apply();
    }

    // Loads tasks from SharedPreferences and reconstructs the task list
    private void loadTasks() {
        SharedPreferences prefs = getSharedPreferences(PREFS_NAME, MODE_PRIVATE);
        String json = prefs.getString(TASKS_KEY, null);
        if (json != null) {
            try {
                JSONArray array = new JSONArray(json);
                for (int i = 0; i < array.length(); i++) {
                    JSONObject obj = array.getJSONObject(i);
                    Task task = addTask(obj.getString("title"), obj.getString("desc"), obj.getString("color"),
                            obj.optString("category", "Personal"),
                            obj.isNull("dueDate") ? null : obj.getLong("dueDate"),
                            obj.getBoolean("isCompleted"));
                    taskList.add(task);
                }
            } catch (JSONException e) { e.printStackTrace(); }
        }
    }
}
```

Activity Registration :

```
public class RegisterActivity extends AppCompatActivity {

  private static final String PREFS_NAME = "TasksPrefs";
  private static final String USER_NAME_KEY = "user_name";
  private static final String IS_REGISTERED_KEY = "is_registered";

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    SharedPreferences prefs = getSharedPreferences(PREFS_NAME, MODE_PRIVATE);
    if (prefs.getBoolean(IS_REGISTERED_KEY, false)) {
      startActivity(new Intent(this, MainActivity.class));
      finish();
      return;
    }

    setContentView(R.layout.activity_register);

    EditText nameInput = findViewById(R.id.nameInput);
    Button startBtn = findViewById(R.id.startBtn);

    startBtn.setOnClickListener(v -> {
      String name = nameInput.getText().toString().trim();
      if (name.isEmpty()) {
        Toast.makeText(this, "Please enter your name", Toast.LENGTH_SHORT).show();
      } else {
        prefs.edit()
          .putString(USER_NAME_KEY, name)
          .putBoolean(IS_REGISTERED_KEY, true)
          .apply();

        startActivity(new Intent(this, MainActivity.class));
        finish();
      }
    });
  }
}
```

XML Code :

XML Main:

```
<TextView
  android:id="@+id/title"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:layout_marginStart="20dp"
  android:layout_marginTop="20dp"
  android:text="My Todos"
  android:textColor="@color/black"
  android:textSize="22sp"
  android:textStyle="bold"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toTopOf="parent" />

<ImageButton
  android:id="@+id/resetBtn"
  android:layout_width="40dp"
  android:layout_height="40dp"
  android:layout_marginTop="16dp"
  android:layout_marginEnd="16dp"
  android:background="?attr/selectableItemBackgroundBorderless"
  android:padding="8dp"
  android:src="@android:drawable/ic_menu_revert"
  android:contentDescription="Reset App"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintTop_toTopOf="parent"
  app:tint="@color/gray_text" />

<EditText
  android:id="@+id/search"
  android:layout_width="0dp"
  android:layout_height="45dp"
  android:layout_marginHorizontal="20dp"
  android:layout_marginTop="15dp"
  android:background="@drawable/search_bg"
  android:drawablePadding="10dp"
  android:hint="Search todos..."
  android:paddingHorizontal="15dp"
  android:textSize="14sp"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toBottomOf="@id/title" />
```

```
<HorizontalScrollView
  android:id="@+id/tabsScroll"
  android:layout_width="match_parent"
  android:layout_height="wrap_content"
  android:layout_marginTop="15dp"
  android:scrollbars="none"
  app:layout_constraintTop_toBottomOf="@id/search">

  <LinearLayout
    android:id="@+id/tabs"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:paddingHorizontal="20dp">

    <TextView
      android:id="@+id/tabAll"
      style="@style/CategoryTabStyle"
      android:text="@string/category_all" />

    <TextView
      android:id="@+id/tabPersonal"
      style="@style/CategoryTabStyle"
      android:text="@string/category_personal" />

    <TextView
      android:id="@+id/tabWork"
      style="@style/CategoryTabStyle"
      android:text="@string/category_work" />

    <TextView
      android:id="@+id/tabShopping"
      style="@style/CategoryTabStyle"
      android:text="@string/category_shopping" />

    <TextView
      android:id="@+id/tabIdeas"
      style="@style/CategoryTabStyle"
      android:text="@string/category_ideas" />
  </LinearLayout>
</HorizontalScrollView>

<androidx.core.widget.NestedScrollView
  android:layout_width="match_parent"
  android:layout_height="0dp"
  android:layout_marginTop="10dp"
  app:layout_constraintBottom_toBottomOf="parent"
```

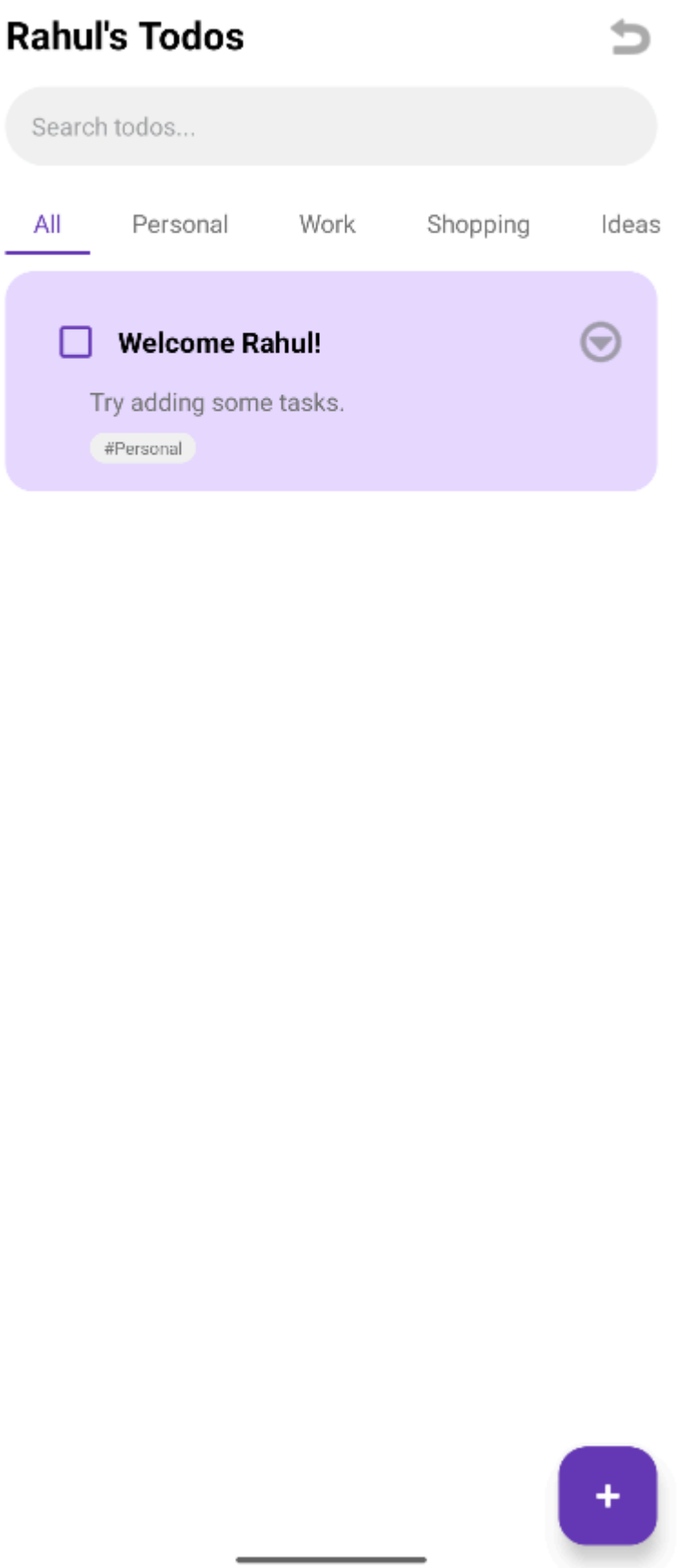
```
  app:layout_constraintTop_toBottomOf="@id/tabsScroll">

  <LinearLayout
     android:id="@+id/taskContainer"
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:orientation="vertical"
     android:paddingHorizontal="20dp"
     android:paddingBottom="80dp" />
</androidx.core.widget.NestedScrollView>

<com.google.android.material.floatingactionbutton.FloatingActionButton
  android:id="@+id/addBtn"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:layout_margin="20dp"
  android:src="@android:drawable/ic_input_add"
  app:backgroundTint="@color/purple_primary"
  app:layout_constraintBottom_toBottomOf="parent"
  app:layout_constraintEnd_toEndOf="parent"
  app:tint="@color/white" />
```

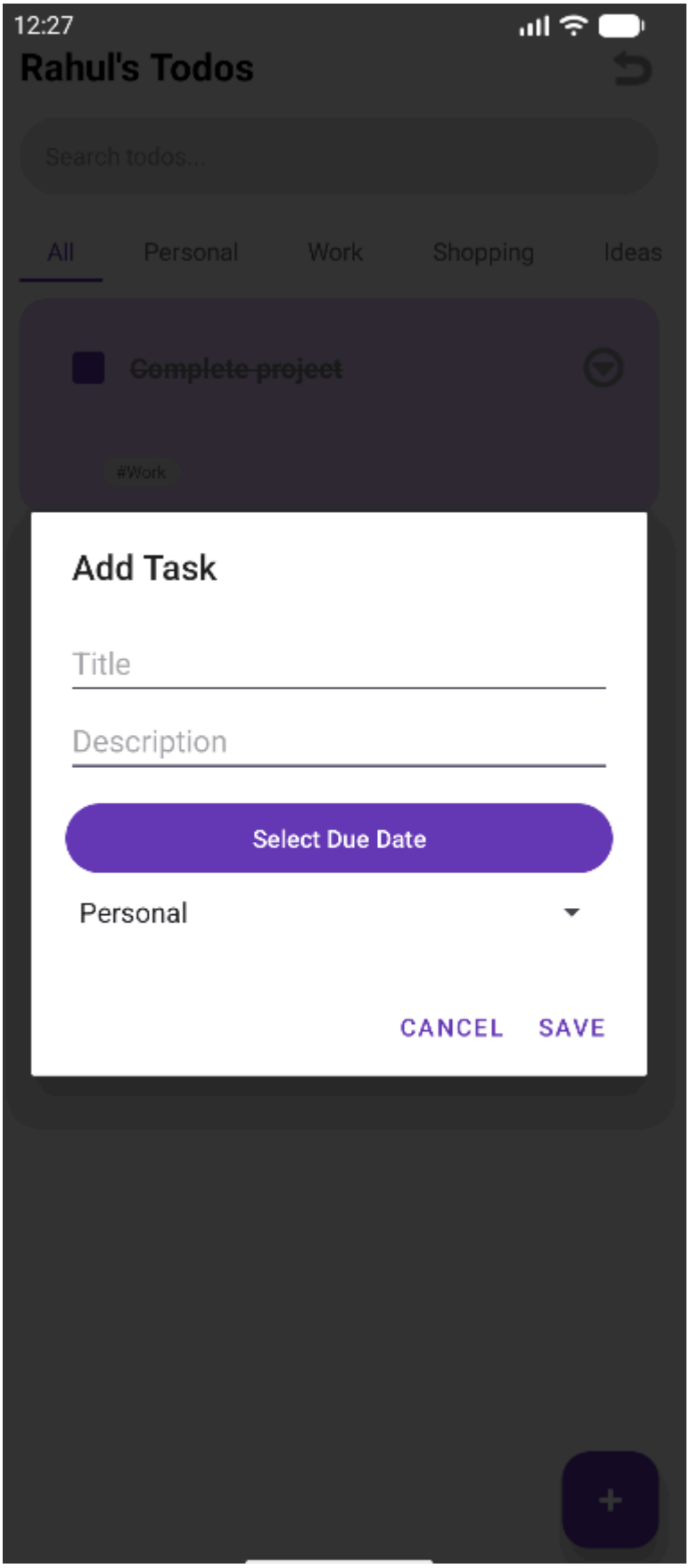
XML Registration :

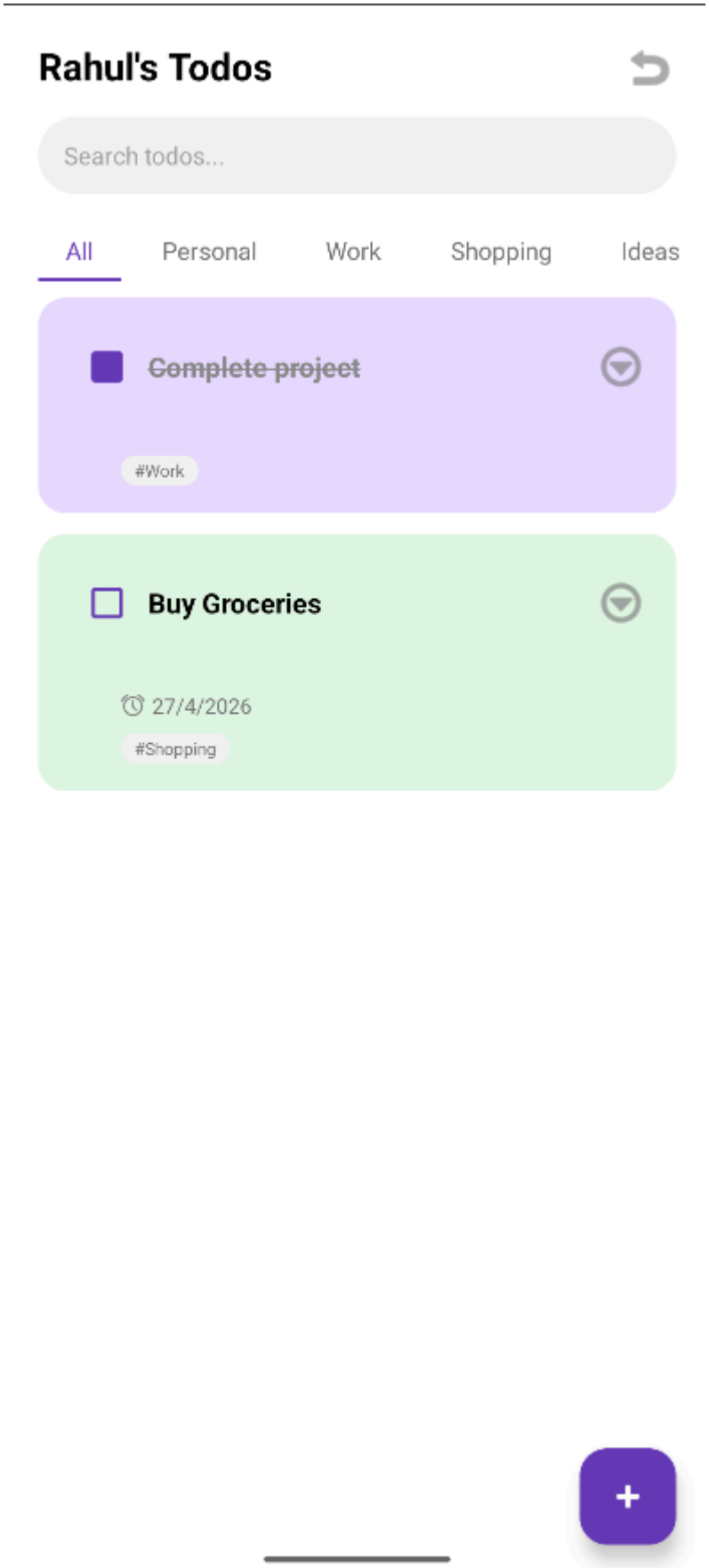
```
<ImageView
  android:id="@+id/logo"
  android:layout_width="120dp"
  android:layout_height="120dp"
  android:layout_marginTop="80dp"
  android:src="@android:drawable/ic_menu_edit"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toTopOf="parent"
  app:tint="@color/purple_primary" />

<TextView
  android:id="@+id/welcomeTitle"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:layout_marginTop="24dp"
  android:text="Welcome to My Todos"
  android:textColor="@color/black"
  android:textSize="24sp"
  android:textStyle="bold"
  app:layout_constraintEnd_toEndOf="parent"
```

```
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/logo" />

<TextView
    android:id="@+id/subtitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="8dp"
    android:text="Please enter your name to continue"
    android:textColor="@color/gray_text"
    android:textSize="16sp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/welcomeTitle" />

<EditText
    android:id="@+id/nameInput"
    android:layout_width="match_parent"
    android:layout_height="56dp"
    android:layout_marginTop="48dp"
    android:background="@drawable/search_bg"
    android:hint="@string/enter_name_hint"
    android:inputType="textPersonName"
    android:paddingHorizontal="20dp"
    app:layout_constraintTop_toBottomOf="@id/subtitle" />

<Button
    android:id="@+id/startBtn"
    android:layout_width="match_parent"
    android:layout_height="56dp"
    android:layout_marginTop="24dp"
    android:text="Get Started"
    android:textAllCaps="false"
    android:textSize="18sp"
    android:backgroundTint="@color/purple_primary"
    android:textColor="@color/white"
    app:layout_constraintTop_toBottomOf="@id/nameInput" />
```

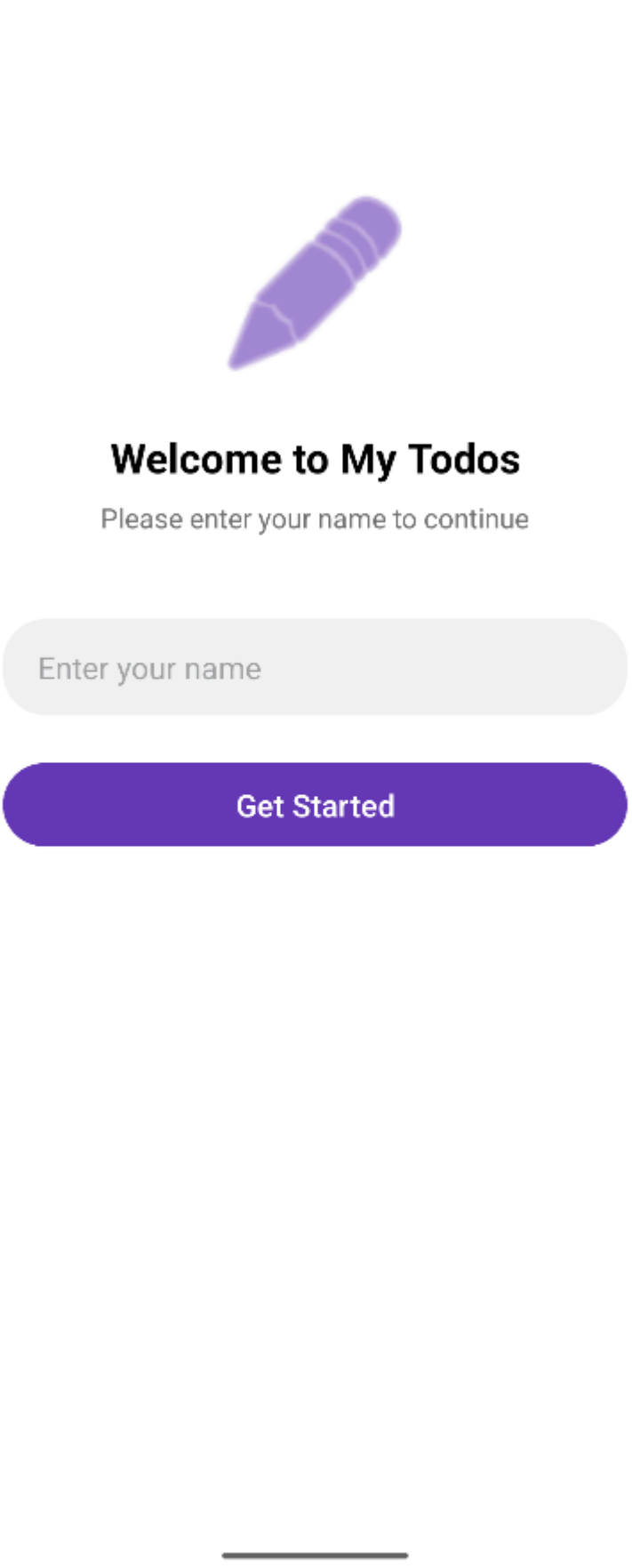
Screenshots :



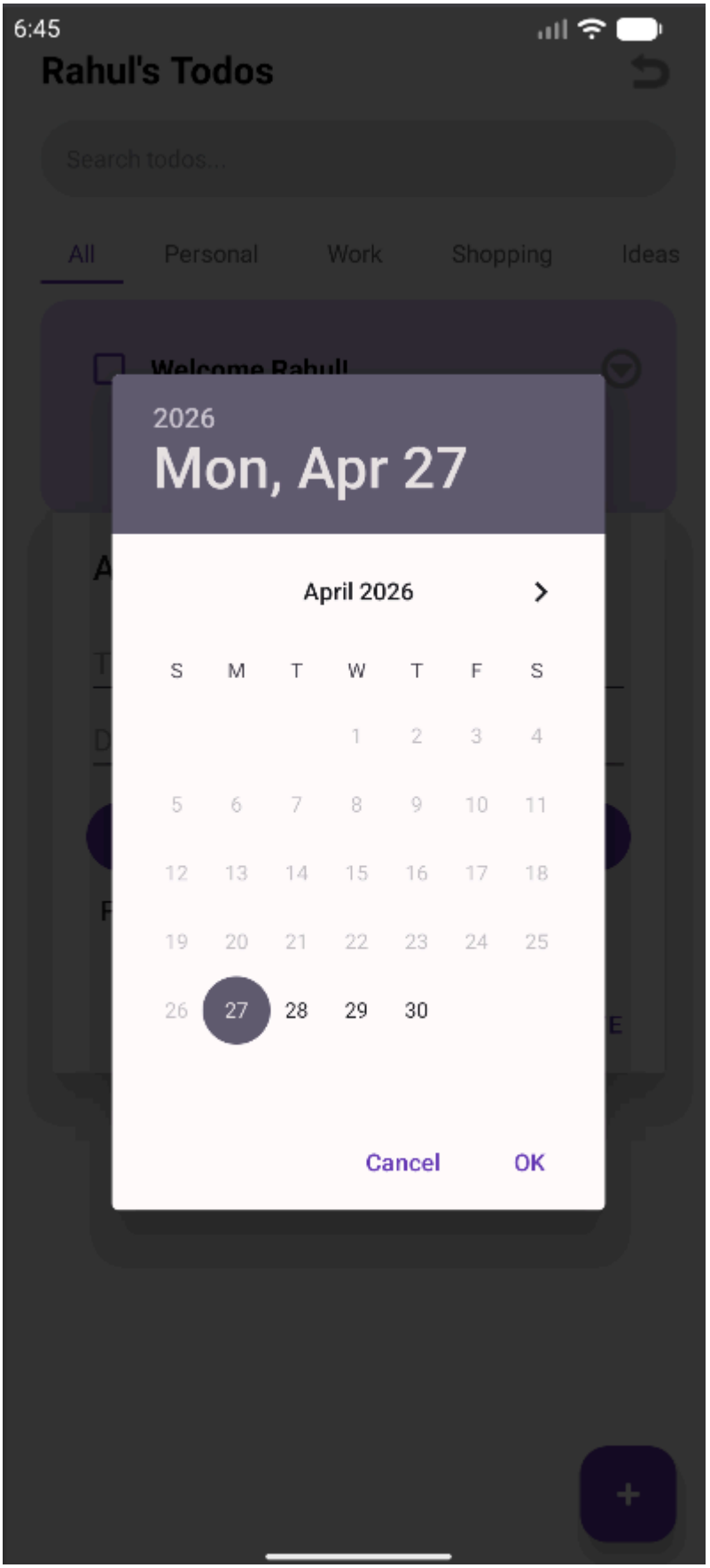

12:27
Rahul's Todos
Search todos...
All
Personal
Work
Shopping
Ideas
Complete project
#Work
Add Task
Title
Description
Select Due Date
Personal
CANCEL
SAVE



Rahul Sehraya (25/SCA/MCAN/081)



Rahul Sehraya (25/SCA/MCAN/081)


6:45
Rahul's Todos
Search todos...
All
Personal
Work
Shopping
Ideas
2026
Mon, Apr 27
April 2026
S M T W T F S
1 2 3 4
5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30
Cancel
OK